

Akshay Borse

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years of experience designing large-scale distributed systems, AI-powered services, and security governance platforms across Microsoft and Amazon. Delivered systems supporting 34K+ enterprise tenants, 25M+ actors, and 100K+ TPS workloads while driving \$28M+ profit generation and \$250K/month cost savings.

EXPERIENCE

Senior Software Engineer | Microsoft | Redmond, WA | Sep-2025 to Present

(C#, .NET, Azure Functions, Cosmos DB, ARM, Bicep, Agentic AI, Agentic Risk, Microsoft Purview, Event Hub, Geneva, Kusto, Data Governance)

- Built Agent Adaptive Protection, enabling tenant-configurable risk scoring and automated policy enforcement for **237K+ AI agents** across **13K+ enterprise tenants**.
- Built AI-agent historical enrichment infrastructure for Microsoft Purview Insider Risk Management, automatically backfilling **90 days of audit history** across **40+ global forests** and **3 sovereign clouds**.
- Directed phased rollout of AI-agent risk capabilities across GCC, GCCH, and DoD sovereign clouds, establishing AI agents as first-class risk actors for enterprise and government tenants.
- Built an isolated AI-agent ingestion platform across 28 regions, processing **135K+ daily risk signals** while preventing agent workloads from impacting human-user processing.

Software Engineer 2 | Amazon | Seattle, WA | Aug 2019 – Aug 2025

(Java, Python, Rest, gRPC, Docker, ECS, Dynamo, S3, Lambda, Aurora, Step Function, App Mesh, Redis, Kibana, OpenSearch, Kinesis, CloudWatch, SageMaker, Bedrock, Anthropic's LLM, Prompt Engineering)

- Designed a microservice bootstrap library that standardized service creation across teams—**cutting setup time from weeks to hours and saving 40–50 weeks**. Leveraged custom DSL and enabled plug-and-play support for AWS infra using declarative config.
- Enhanced the bootstrap library for Multi-Service Deployment in a single pipeline, optimizing maintenance for a microservices architecture with 60+ services. **Reduced the service count by 9**, with plans in place to further streamline and drive operational excellence.
- Optimized Docker builds, cutting container image size by 83% and build time by over 50%, accelerating CI/CD and saving 60GB+ of storage.
- Established and led a logging/metrics optimization initiative and process during AWS migration—standardized logs, removed redundant code, cutting noise, improving debuggability, and **reducing CloudWatch costs by \$250K+/month**.
- Built a Restock Alerting Service to notify customers when out-of-stock items were restocked increasing sales by 13% for cancelled ASIN's.
- Implemented the Just-In-Stock service on native AWS to track real-time inventory changes and auto-optimize fulfillment, increasing early delivery rates by 16% and improving supply chain responsiveness.
- Designed a self-service UI and backend to modify delivery dates for high-volume Amazon Business orders—**eliminating 2–3 hours of manual ops per request** and reducing dependency on external teams. Also mentored an intern during development.
- Engineered a Replay Service using a lightweight wrapper over AWS Step Functions to schedule event replays up to 1 year—eliminating cron-based scheduling and reducing operational overhead with built-in visibility and fault tolerance.
- Configured a **dynamic log-level API for production**, enabling real-time debugging without redeployments and cutting incident resolution time by over an hour—especially valuable in complex, distributed services.
- Led the design for a RAG-based service that leverages a model deployed on Amazon SageMaker to integrate real-time inventory signals and vector search, improving delivery predictions and driving a **7% increase in global sales**.
- Redesigned 37 email templates by integrating an LLM via Amazon Bedrock, using a RAG-based approach with a curated corpus to constrain generation—ensuring compliance, improving personalization, and reinforcing brand consistency.
- Integrated a binary classification model (XGBoost) to predict potential order delays before SLA breach, enabling early delay alerts to customers and improving trust in delivery communication.
- Architected and deployed backend integration infrastructure for SageMaker hosted quantile regression model powering real-time delivery range predictions, driving statistically significant annualized **~28MM profit gain**.
- Architected an event orchestration layer for AWS-based microservices, enabling real-time coordination across 100K+ TPS and 16 services—balancing REST and gRPC protocols to support 8 business-critical workflows.
- Coordinated **end-to-end load testing for 60+ distributed services** during high-traffic events (Prime Day, Black Friday), aligning with upstream and downstream service owners to proactively surface scaling issues and ensure fault tolerance under peak load.
- Developed a dry-run simulation tool for incoming orders, allowing non-engineering stakeholders to visualize system behavior—**eliminating engineering dependencies, improving planning agility, and reducing deep-dive turnaround time from hours to minutes**.
- Proactively led cultural and operational improvements by launching on-call reduction reviews, post-launch knowledge sharing, and design discussion sessions—reducing Sev2 alerts by 60%, improving team-wide system understanding, and strengthening long-term design thinking.
- Led TRA readiness for a Tier 1 service (CI/CD, multi-AZ, fault injection, canary), while building reusable tools to improve dev efficiency—automated UML sync to wiki, introduced reusable wiki headers, and reduced test setup complexity via Lugia integration.

Software Engineer | Liquiron | San Jose, CA | Dec-2018 to Jan-2019

(OAuth 2.0, Passport.js, Rest API, Vue.js, Google Firebase, Node.js, Mocha & Chai, Jest, Serverless Hosting, Stateless Backend)

- Re-architected authentication using OAuth 2.0 and Passport.js, improving security, modularity, and maintainability.
- Migrated backend services to Firebase serverless infra & developed REST APIs supporting SPA workflows, reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India | Sept-2016 to Jul-2017

(CNN, AWS EC2, REST, NodeJS, JavaScript, Python, Axios, Quasar)

- Developed a real-time **CNN-based sentiment classification system** achieving 87% accuracy across seven sentiment categories.
- **Modernized legacy web applications into a single-page architecture**, improving response time by 25% and offline performance by 9%, while enhancing developer productivity through a feedback system adopted by six engineering teams.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. | Sep - 2017 to Jun-2019 | GPA - 3.77

- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. | Jun - 2012 to Jun-2016 | GPA - 3.34

SELECTED ENGINEERING PROJECTS - [HTTPS://GITHUB.COM/AMSBORSE](https://github.com/amsborse)

- Developed LeetDesign, a distributed systems simulator that transforms architecture diagrams into executable models for evaluating scalability, resilience, and failure trade-offs.
- Designed Finance OS, a cloud-native financial intelligence platform that combines deterministic rule engines and LLM-powered analysis to convert raw financial data into actionable insights. <https://finance-os-lilac.vercel.app/>
- Built Invisible Loops, an interactive learning platform that helps users recognize and resolve recurring cognitive and emotional patterns through narrative-driven exploration and immersive visualization. <https://invisible-loops-two.vercel.app/>